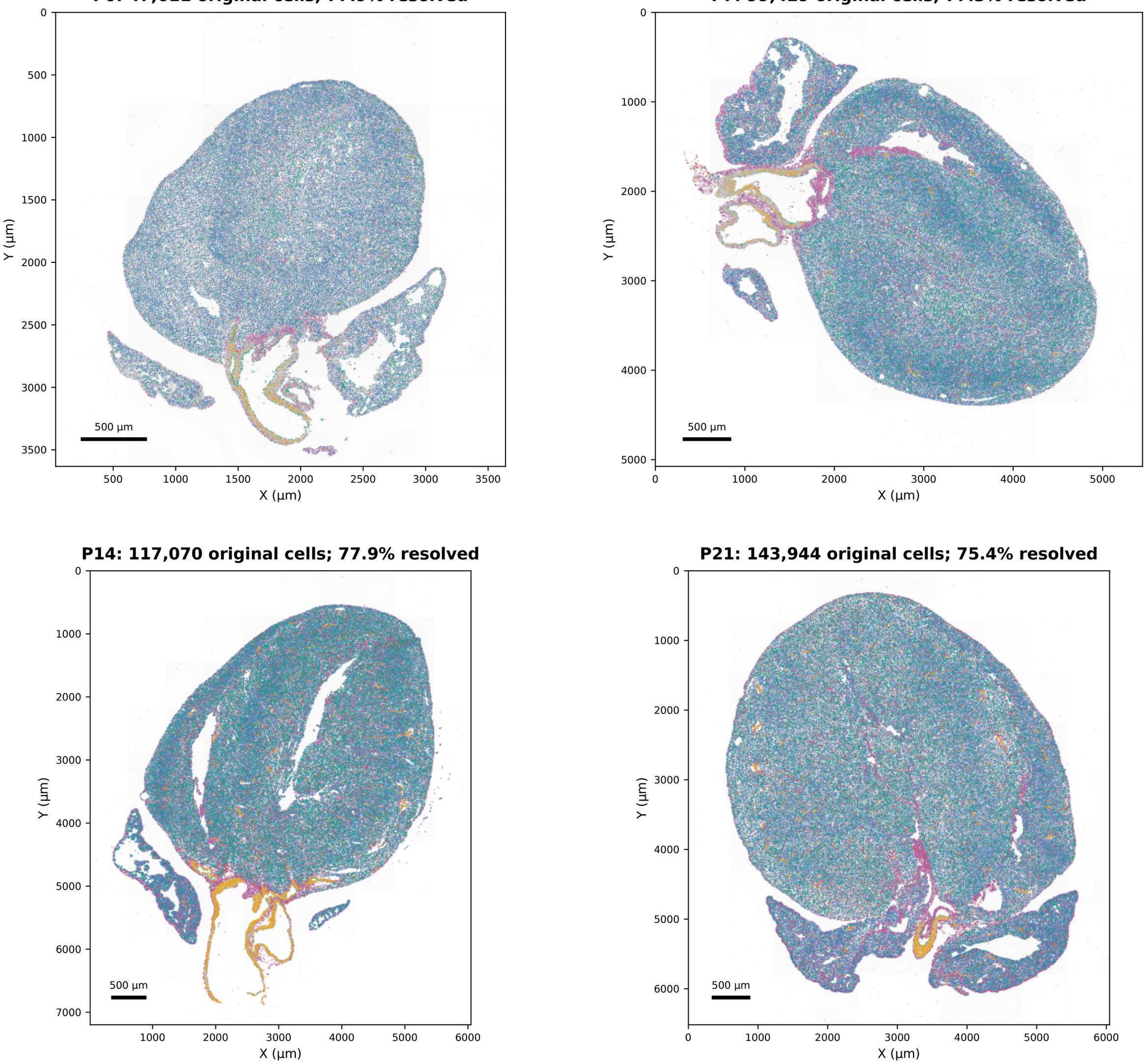
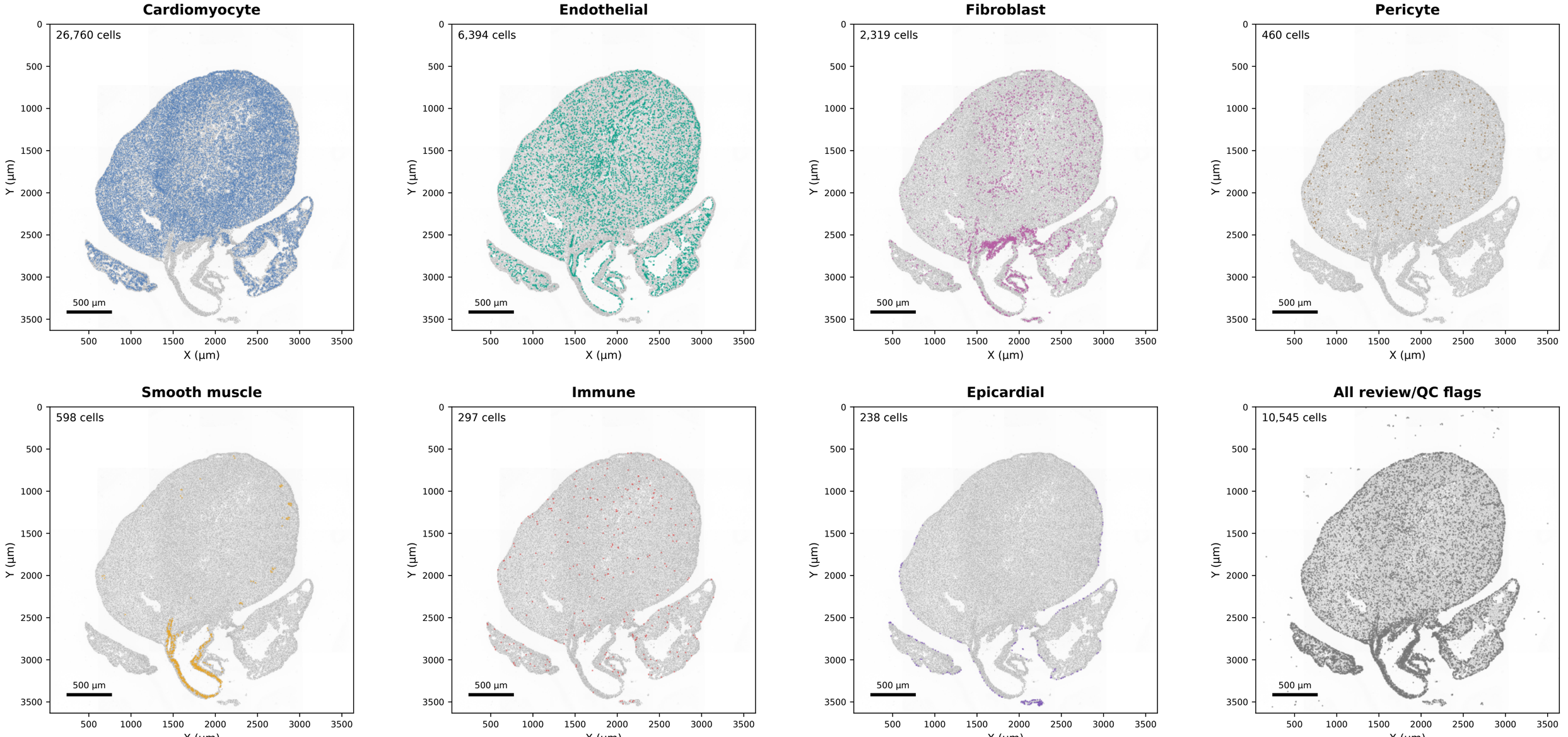
All cardiac cells on the native P0–P21 Xenium sections

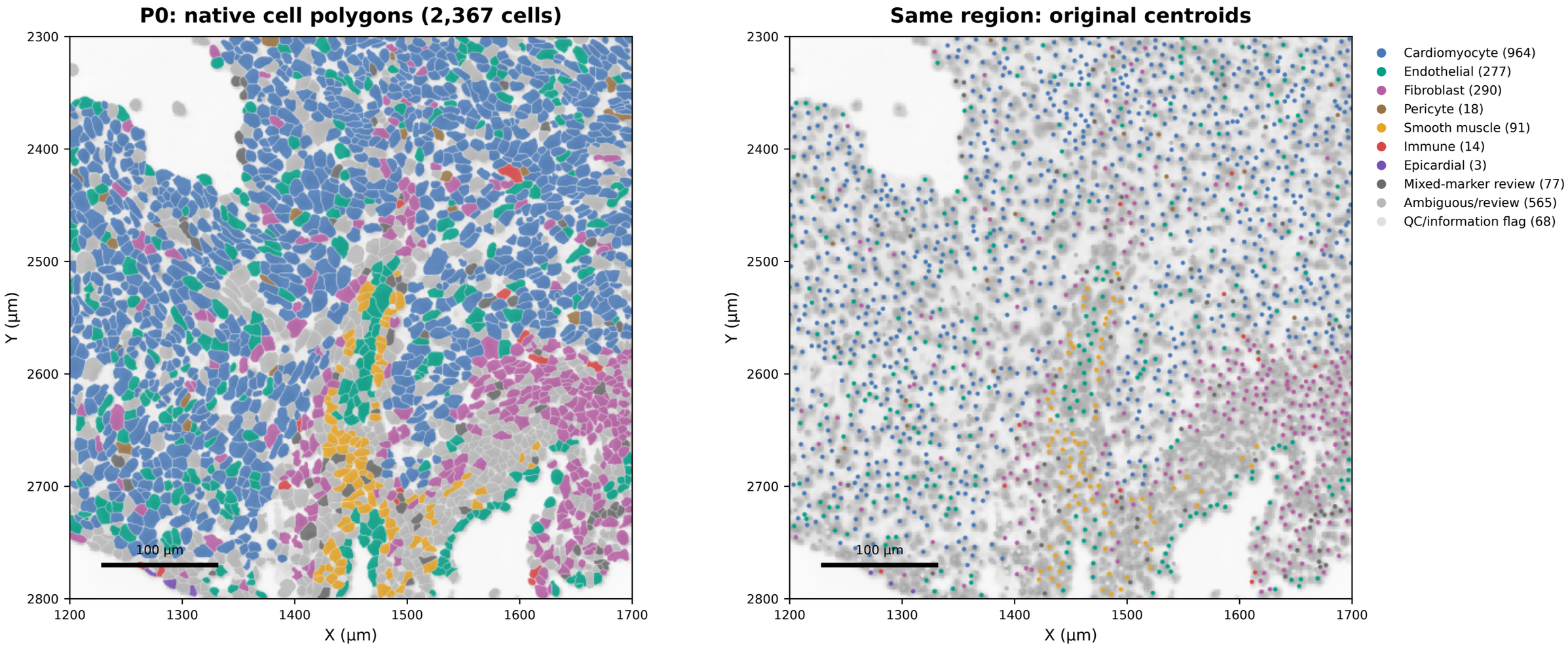


Same colors across stages. Direct broad-lineage assignments; uncertain and mixed profiles remain gray.
Each panel is its own acquired section, shown in native µm coordinates; sections are not registered to one another.

P0: spatial distributions of individual cardiac lineages

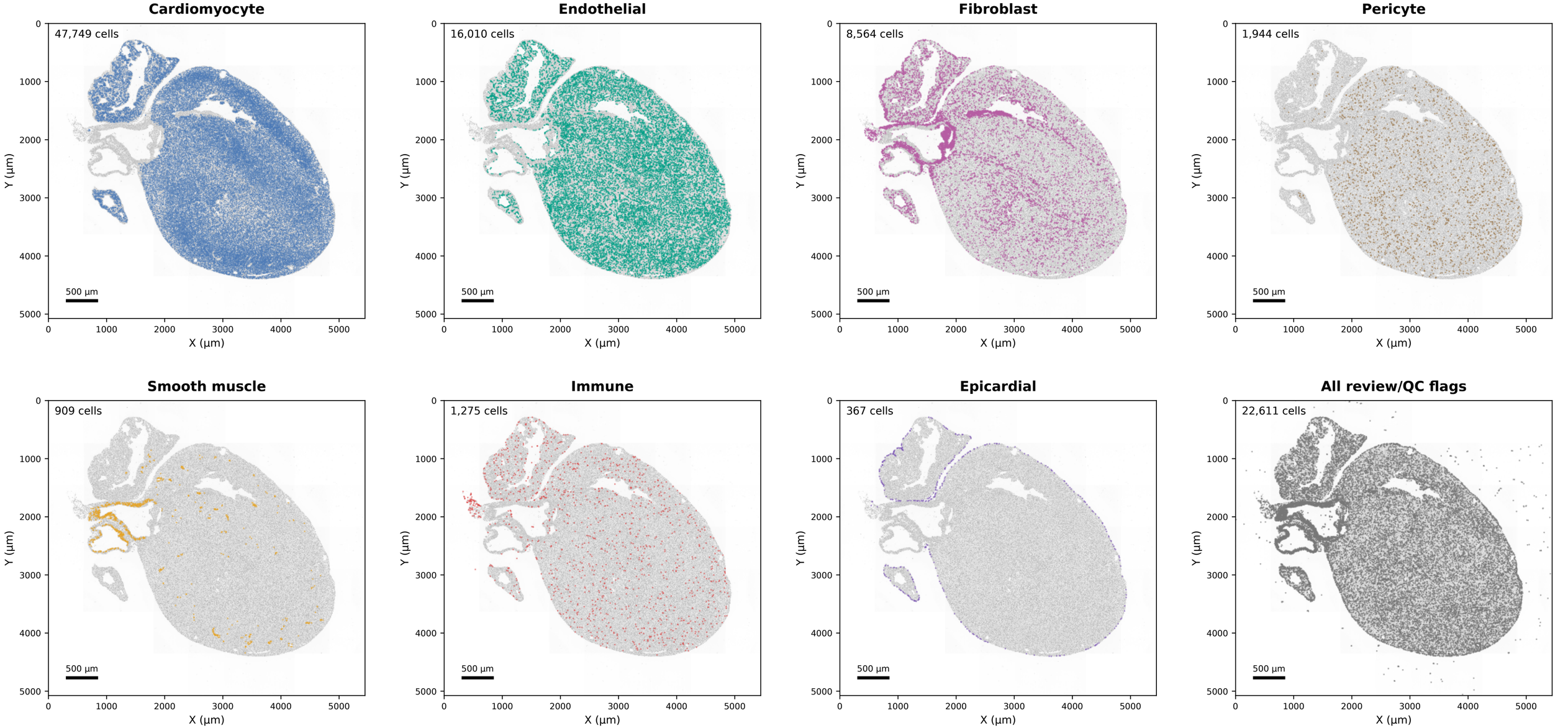


Each panel highlights one lineage on the same native section. Other original cells are light gray.
No assigned glial category is shown because the reference glial-marker panel is unassayed; this is not evidence of biological absence.

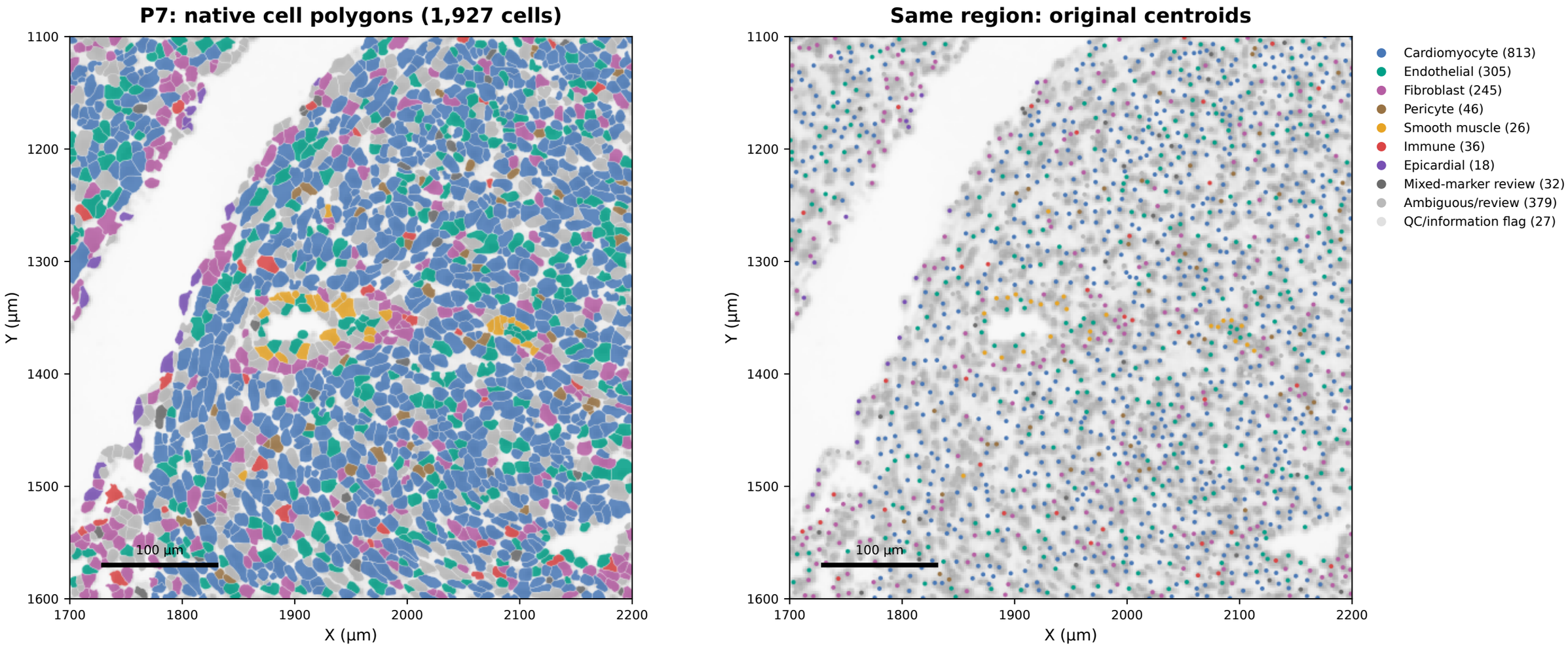


Example selected by high broad-lineage diversity in a 100 µm tile (≥40 resolved cells and ≥4 lineages when available).
Both panels show the same 500 × 500 µm region; no cell positions are moved. Cell segmentation is deposited, not newly inferred.

P7: spatial distributions of individual cardiac lineages

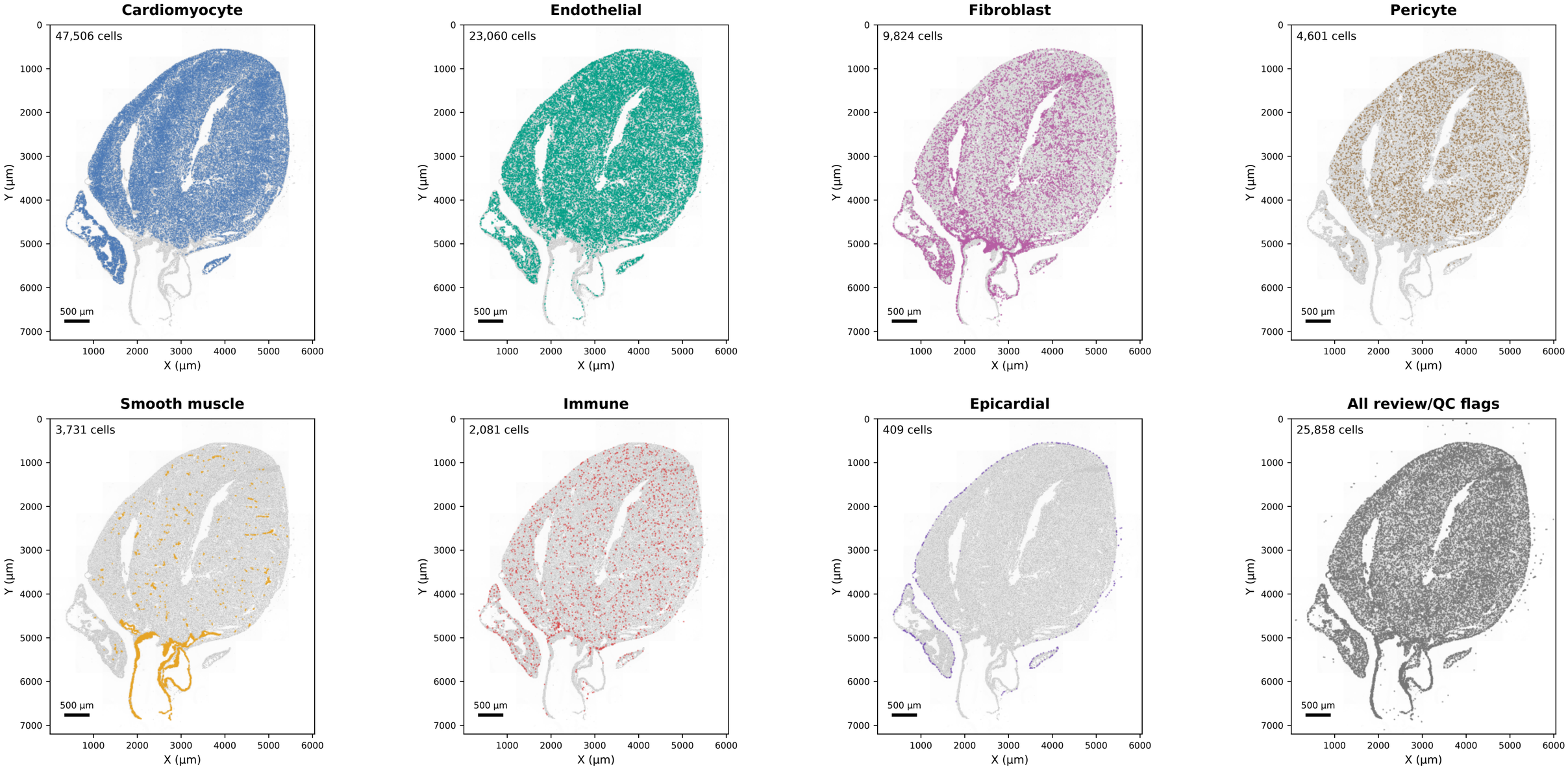


Each panel highlights one lineage on the same native section. Other original cells are light gray.
No assigned glial category is shown because the reference glial-marker panel is unassayed; this is not evidence of biological absence.

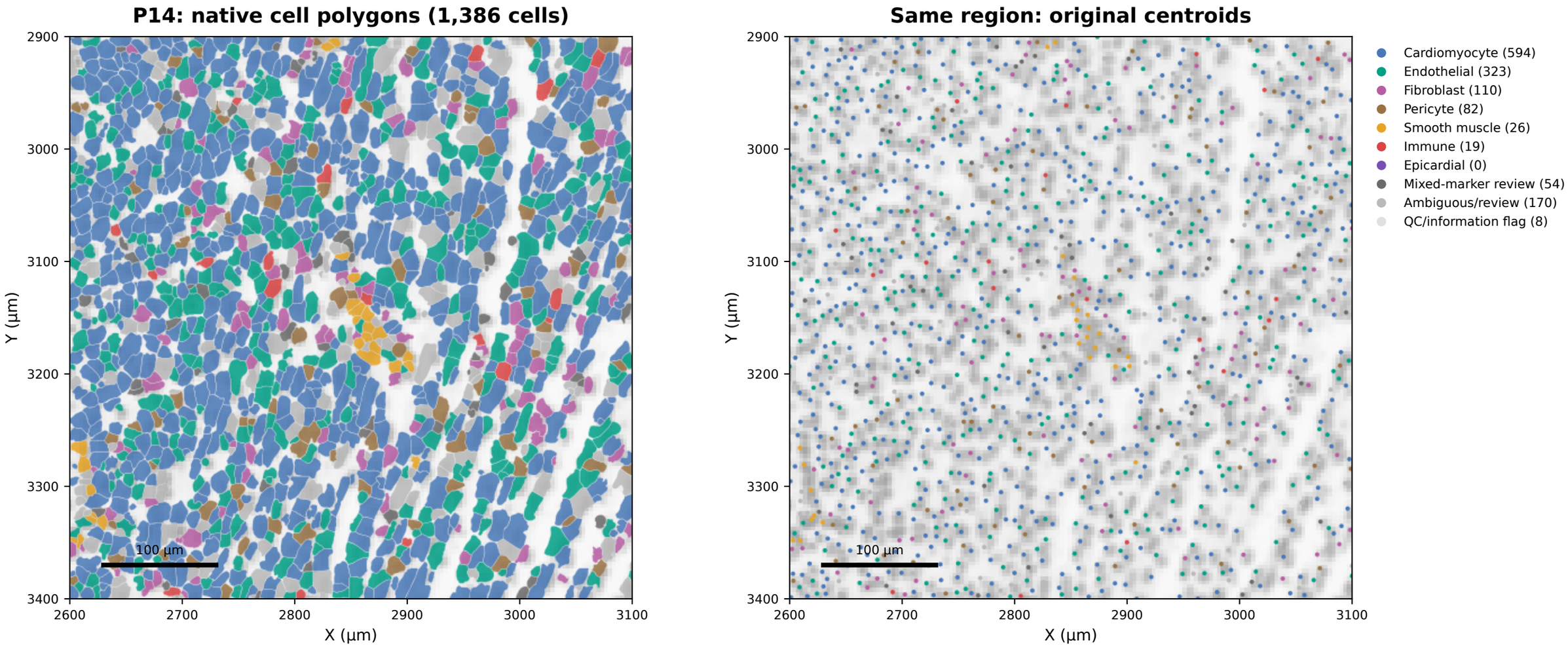


Example selected by high broad-lineage diversity in a 100 µm tile (≥40 resolved cells and ≥4 lineages when available).
Both panels show the same 500 × 500 µm region; no cell positions are moved. Cell segmentation is deposited, not newly inferred.

P14: spatial distributions of individual cardiac lineages

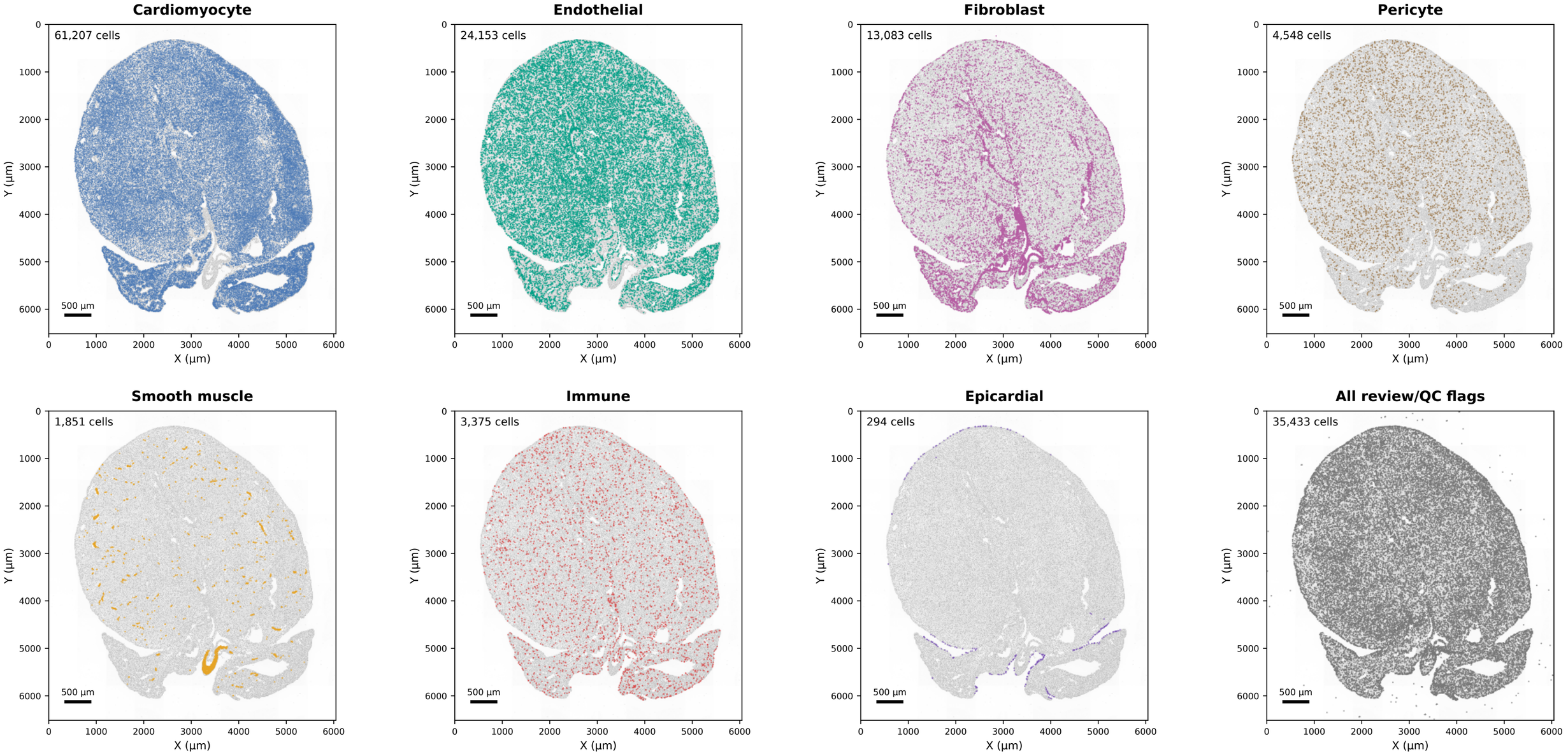


Each panel highlights one lineage on the same native section. Other original cells are light gray.
No assigned glial category is shown because the reference glial-marker panel is unassayed; this is not evidence of biological absence.

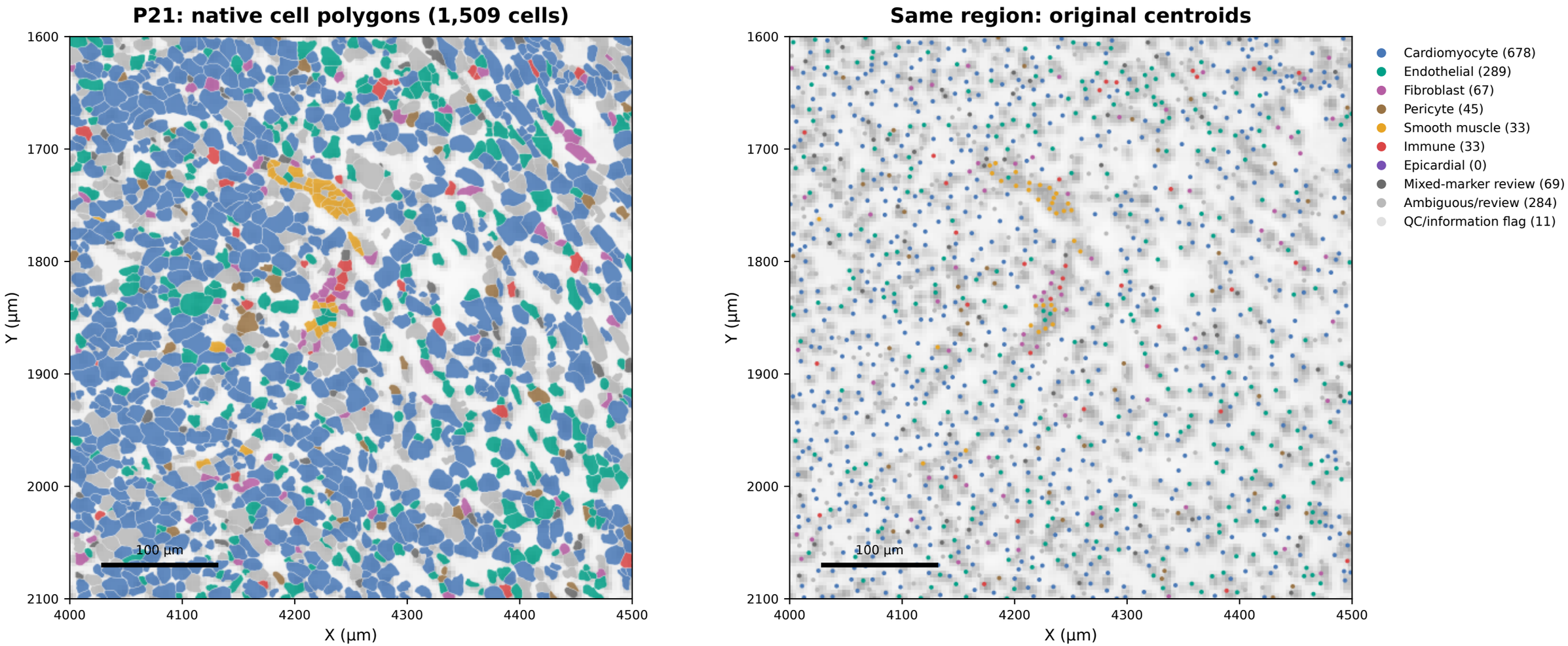


Example selected by high broad-lineage diversity in a 100 µm tile (≥40 resolved cells and ≥4 lineages when available).
Both panels show the same 500 × 500 µm region; no cell positions are moved. Cell segmentation is deposited, not newly inferred.

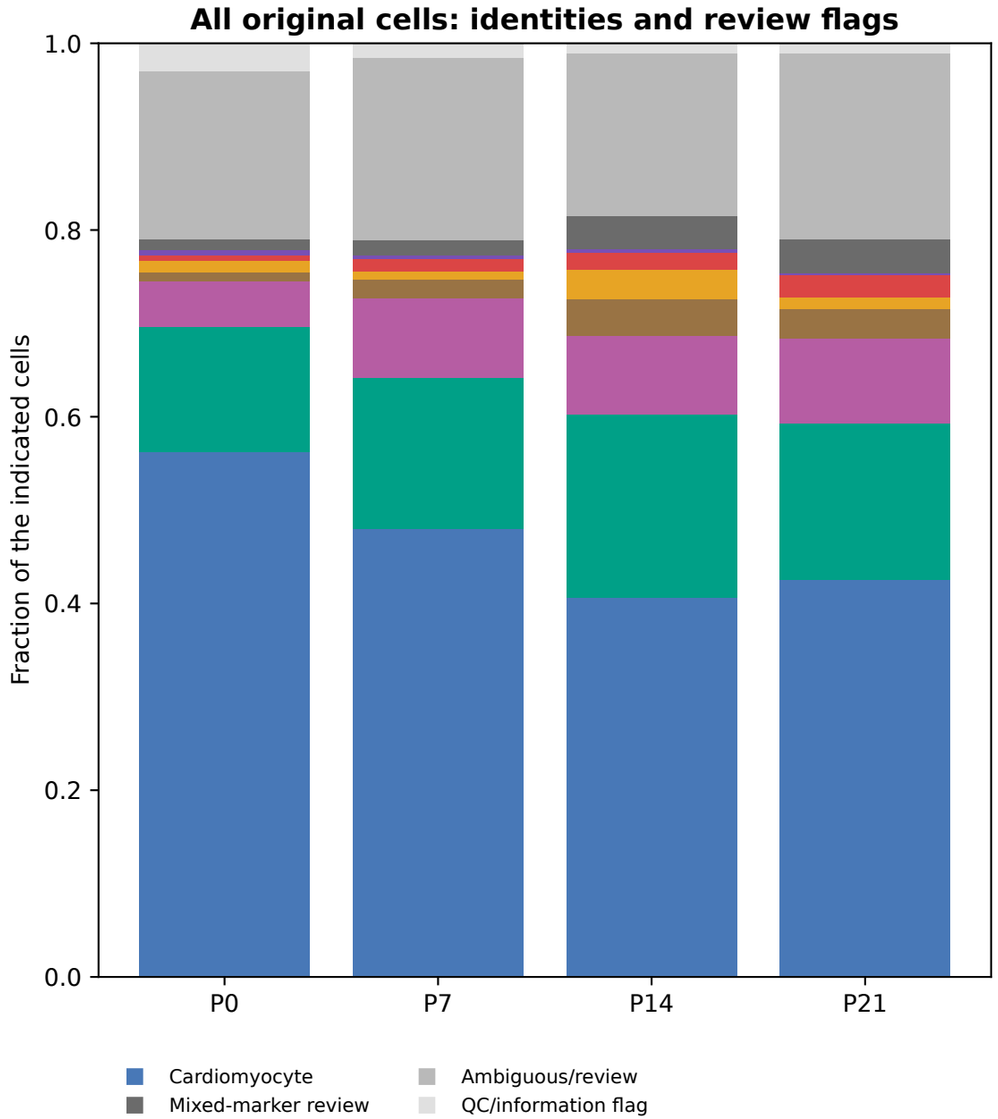
P21: spatial distributions of individual cardiac lineages

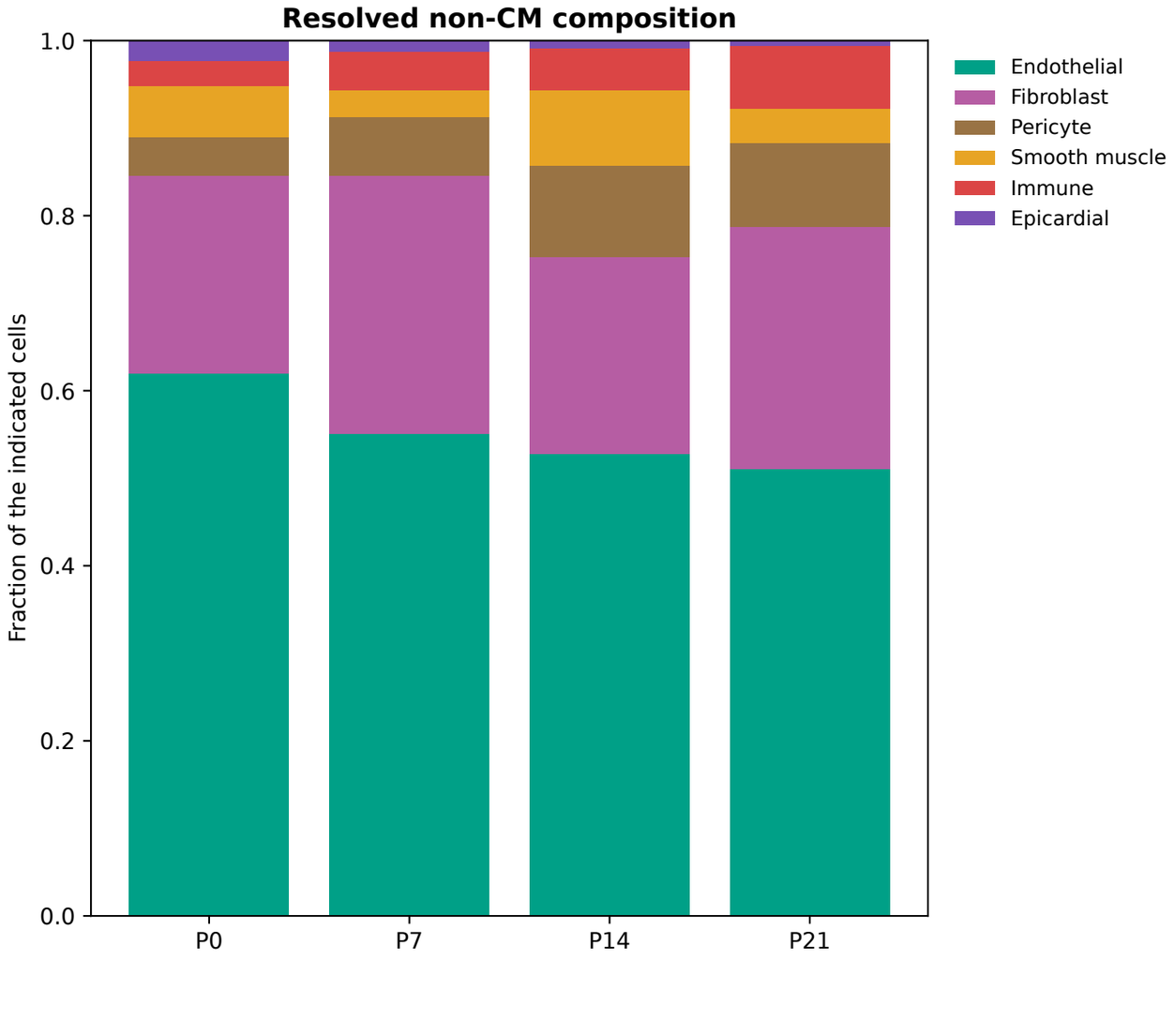


Each panel highlights one lineage on the same native section. Other original cells are light gray.
No assigned glial category is shown because the reference glial-marker panel is unassayed; this is not evidence of biological absence.



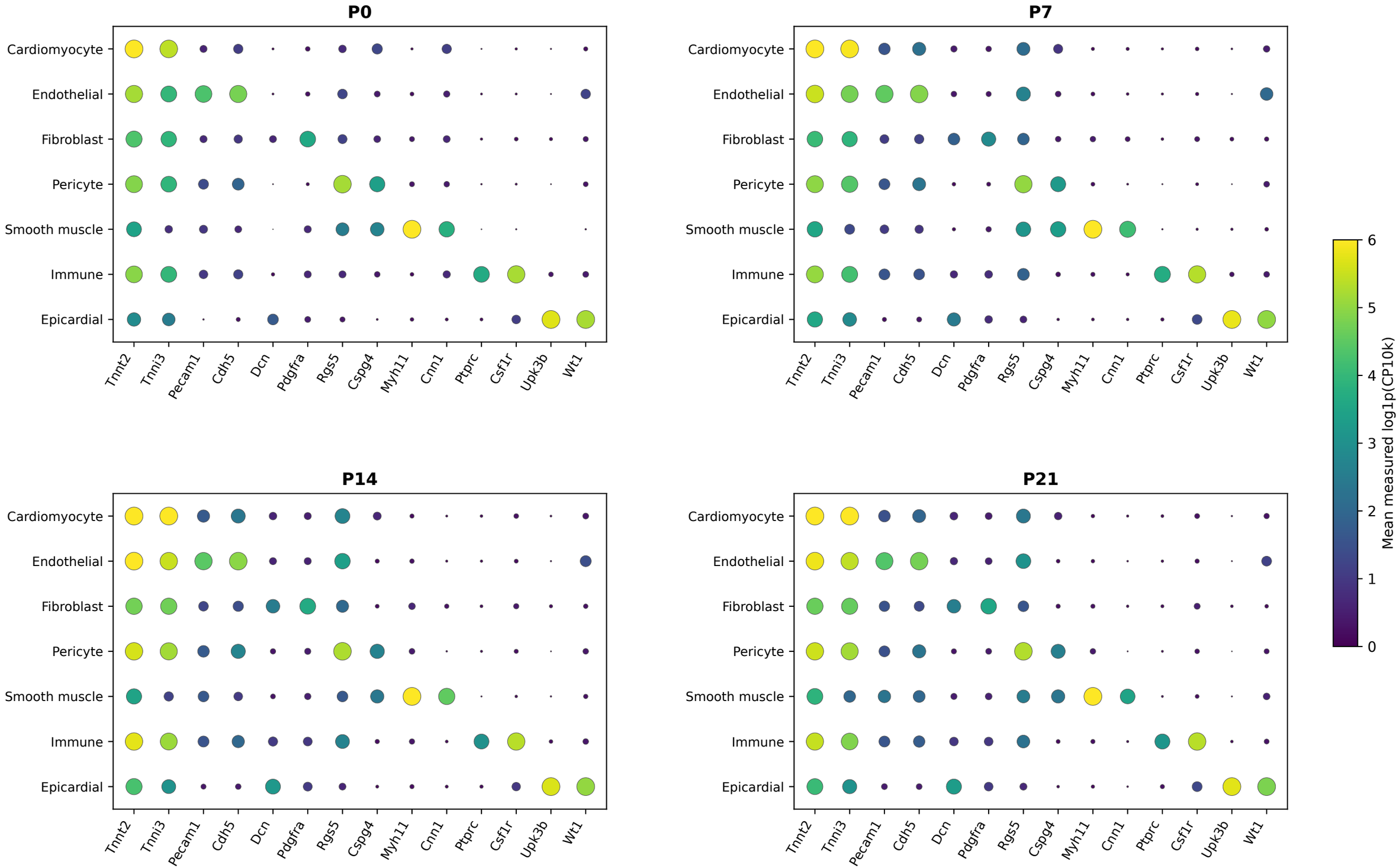
Example selected by high broad-lineage diversity in a 100 µm tile (≥40 resolved cells and ≥4 lineages when available).
Both panels show the same 500 × 500 µm region; no cell positions are moved. Cell segmentation is deposited, not newly inferred.





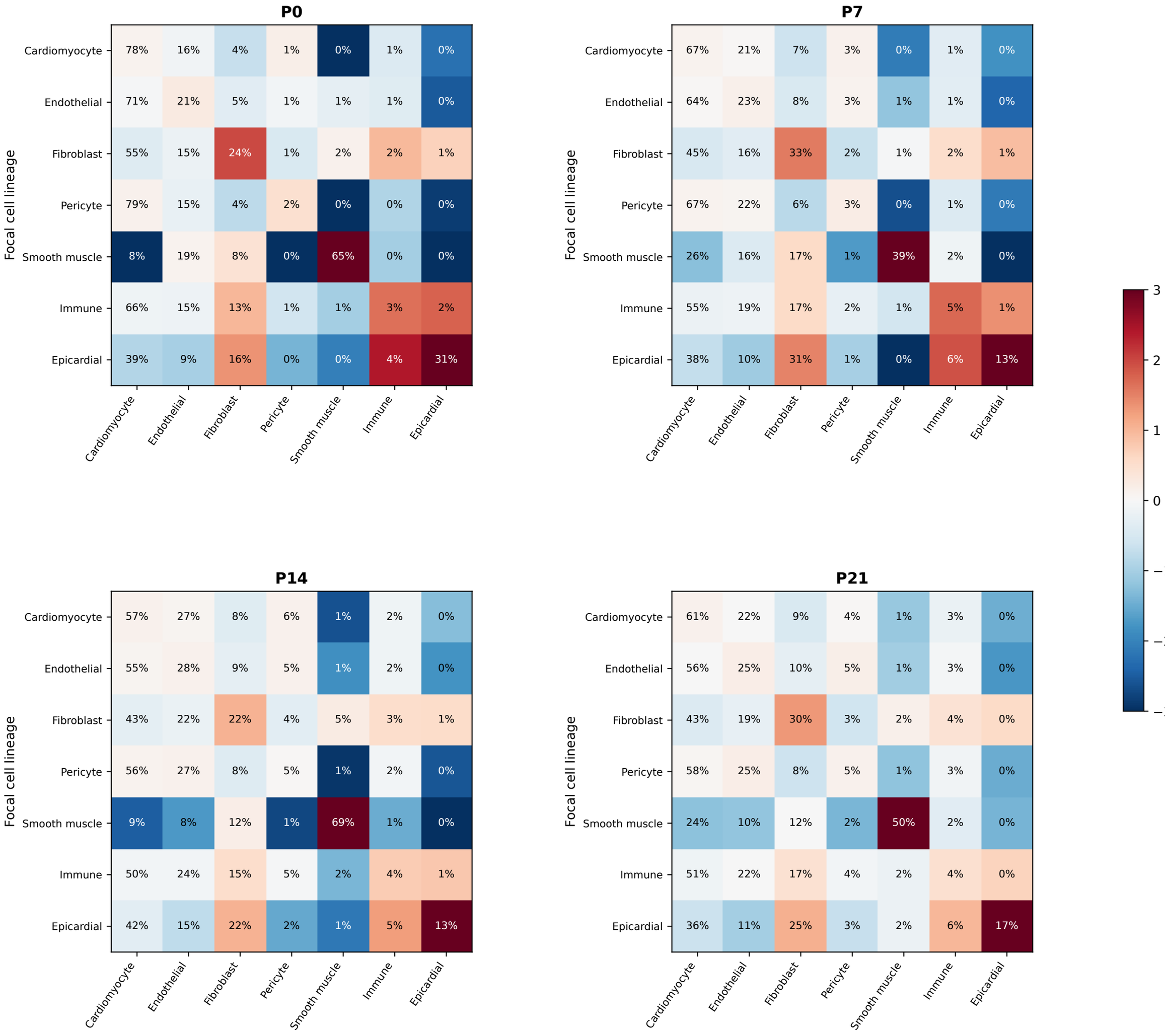
Composition is conditional on the deposited sections, segmentation and conservative assignment criteria.
One section per age; proportions are descriptive and do not estimate replicated developmental changes in cell abundance.

Measured marker consistency of the audited broad lineages



Dot area: detected fraction (0–100%); color: mean measured expression, all403-target normalization.
These genes contribute to annotation and are not independent validation markers. Full gene summaries are downloadable.

Broad-lineage spatial neighborhoods



Text: resolved-neighbor composition. Color: enrichment relative to each section’s overall resolved composition.
20 nearest other original cells within100 µm; focal cells require≥5 resolved neighbors.
Descriptive spatial co-occurrence; cell density, anatomy and ambiguity affect these values. No interaction or significance claim.